Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. We trained each model for 100 epochs with a batch size of 64.

2 Results

Figure 1 shows the validation accuracy of LSTM by dataset.

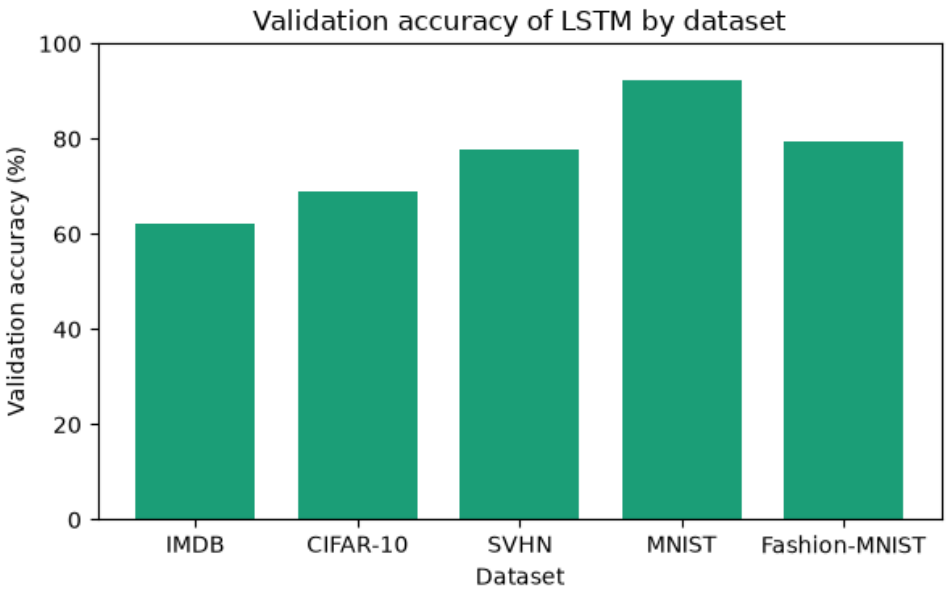

Figure 1: Validation accuracy of LSTM by dataset for the models in this experiment.

On MNIST, LSTM obtains a validation accuracy of 97.2%. LSTM achieves 77.8% validation accuracy on SVHN (Figure 1). As shown in Figure 1, LSTM scores 7.1 points higher on CIFAR-10 than on IMDB. As shown in Figure 1, LSTM achieves 27.8% validation accuracy on SVHN.

Figure 2 shows the validation accuracy by dataset.

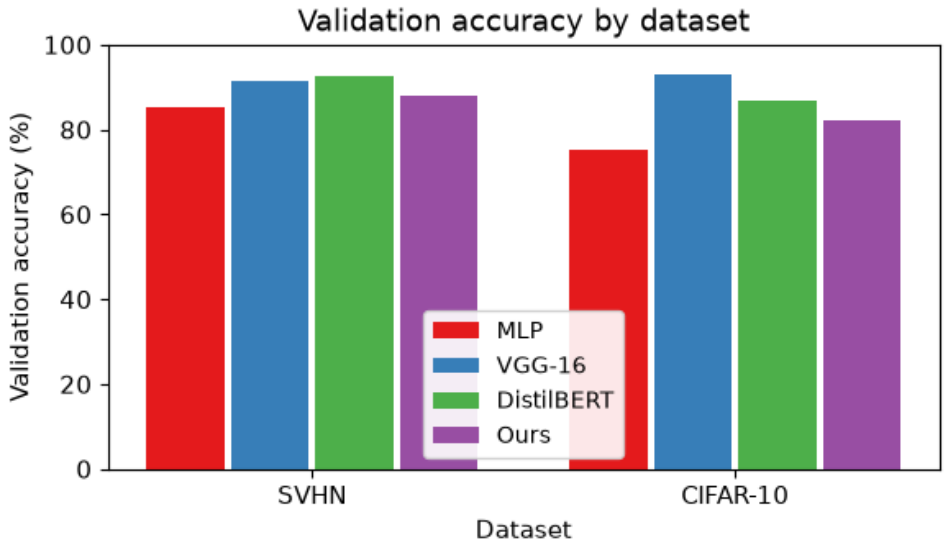

Figure 2: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 2, VGG-16 outperforms MLP by about 18 percentage points on CIFAR-10. On SVHN, Ours is 2.6 points better than MLP (Figure 2). On CIFAR-10, DistilBERT is about 5 points better than Ours (Figure 2). MLP achieves 65.2% validation accuracy on CIFAR-10 (Figure 2).

3 Discussion

Random Forest reaches 81.1% accuracy on MNIST. Overall, the results are in line with what we expected from the literature.